

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think> Anatomical snuffbox is a region in the hand around the base of the thumb that contains the tendons of the abductor pollicis longus, extensor pollicis brevis, and extensor pollicis longus, and the superficial and deep palmar branches of the radial artery. The anatomical snuffbox is supplied by the superficial palmar arch and digital branches of the radial and ulnar arteries. During a stab wound, damage to the superficial palmar arch or a major branch of the radial artery may cause significant bleeding. </think><answer>Radial artery</answer>

